AN A-MODULE ON A SCHEME WITH QUASI-COHERENT ALGEBRA A IS QUASI-COHERENT OVER A EXACTLY WHEN IT IS QUASI-COHERENT OVER THE STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-quasi-coherence-scalar-restriction is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 1844-1853 and proof span 1855-1893 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The counted statement contains a reference environment holding the author's citation of EGA. The project's own shared preamble (now retained in full, 88 lines, per the 2026-10-12 repair) defines the reference environment as a comment environment, so that citation is not typeset by the project itself either; a bibliography is therefore not attached to this card and none is needed.

Editorial note. The counted proof contains the author's parenthetical '(details omitted)' for the routine point that the module structure on M is compatible with the R-module structure; that phrase is kept verbatim and what it leaves to the reader is a compatibility check, not a step of the main argument.

Allowed standard prerequisites (editorial).

- Local presentations of a quasi-coherent module as the cokernel of a map of direct sums of the structure sheaf or of the algebra, on an open neighbourhood of a point. This is what the two directions of the proof use at proof lines 2074-2078 and 2095-2102. Supplies the exact sequences in terms of which the module is recognised as quasi-coherent over the algebra, respectively over the structure sheaf.
- The tilde construction over an affine scheme: a quasi-coherent algebra on Spec(R) is the tilde of an R-algebra, a quasi-coherent module is the tilde of an R-module, and the tilde functor is exact. Used at proof lines 2086-2092 and again at 2103-2108. Translates the second direction into a statement about an R-algebra A and an R-module M with an A-module structure, and transfers an A-module presentation back to sheaves.

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- Cokernels of maps of quasi-coherent modules on a scheme are quasi-coherent, and exactness of a sequence of modules over an algebra regarded as a sequence of modules over the smaller ring. Used at proof lines 2077-2080 and 2092-2093. Closes the first direction and identifies the compatible structures in the affine chart.

Lemma 0.1. *Let S be a scheme and let $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_S$-algebra. An $\mathcal{A}$-module is quasi-coherent as an $\mathcal{O}_S$-module if and only if it is quasi-coherent as an $\mathcal{A}$-module.*

Proof. Let $\mathcal{F}$ be an $\mathcal{A}$-module. If $\mathcal{F}$ is quasi-coherent as an $\mathcal{A}$-module, then for every $s \in S$ there exists an open neighbourhood U of s and an exact sequence

$$\bigoplus\nolimits_J \mathcal{A}|_U \to \bigoplus\nolimits_I \mathcal{A}|_U \to \mathcal{F}|_U$$

of $\mathcal{A}|_U$-modules. Then this is also an exact sequence of $\mathcal{O}_U$-modules. Hence $\mathcal{F}|_U$ is quasi-coherent as the cokernel of a morphism of quasi-coherent $\mathcal{O}_U$-modules on a scheme. It follows that $\mathcal{F}$ is quasi-coherent as an $\mathcal{O}_X$-module.

Conversely, assume $\mathcal{F}$ is quasi-coherent as an $\mathcal{O}_X$-module. Pick an open affine $\mathrm{Spec}(R) = U \subset S$. We have isomorphisms of $\mathcal{O}_U$-modules $\mathcal{A}|_U \cong \widetilde{A}$ and $\mathcal{F}|_U \cong \widetilde{M}$, for some R-algebra A and some R-module M. The $\mathcal{A}$-module structure on $\mathcal{F}$ translates into an A-module structure on M compatible with the given R-module structure (details omitted). Choose an exact sequence

$$\bigoplus\nolimits_J A \to \bigoplus\nolimits_I A \to M \to 0$$

of A-modules. Since the functor $\widetilde{\ }$ is exact, this produces an exact sequence

$$\bigoplus\nolimits_J \widetilde{A} \to \bigoplus\nolimits_I \widetilde{A} \to \widetilde{M} \to 0$$

of $\widetilde{A}$-modules. This means that $\mathcal{F}$ is quasi-coherent as an $\mathcal{A}$-module. □